

RAHUL BALAJI

📞 585-317-3846 | ✉ contact.rahul.balaji@gmail.com | 📍 Nashville, TN

PROFESSIONAL SUMMARY

Senior Technical Lead & Engineer with 7+ years of experience designing scalable cloud-native ecosystems and high-performance data platforms. Proven track record in leading digital transformations, migrating legacy on-prem systems to GCP/AWS, and pioneering Generative AI integration into Business Intelligence Operations. Expert at bridging the gap between complex software engineering, regulated healthcare data standards (HL7/FHIR), and executive-level business intelligence.

PROFESSIONAL EXPERIENCE

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| Senior Manager Analytics | <u>HCA Healthcare</u> | <i>Nashville, TN, USA</i> | 09/2024 - Present |
| <ul style="list-style-type: none">Improved surgical operations visibility by developing and deploying the Surgical Services Power BI dashboard, [2nd Most Utilized Dashboard in HCA] for critical clinical metrics, enabling <u>Surgical Services Leadership</u> to track on-time case starts and optimize resource utilization.Pioneered the adoption of AI-assisted coding (GitHub Copilot) and 'Vibe Coding' workflows; led team training sessions to leverage Generative AI for SQL and Python development, resulting in increased delivery speed and the successful procurement of a ChatGPT Enterprise license for the Analytics Team.Increased reporting accuracy and performance by engineering integrated data pipelines between Google BigQuery and SQL Server using SSIS, Python, Google Cloud Storage, and Cloud Run delivering reliable Power BI performance indicators for the Performance Improvement department.Modernized analytics infrastructure by migrating a 500+ daily-user Anesthesia dashboard from an on-prem SSAS backend to Power BI Service with BigQuery, improving refresh speed by ~30% and reducing maintenance overhead. | | | |
| Data Science Internship | <u>Gap Intelligence</u> | <i>San Diego, CA, USA</i> | 06/2024 - 09/2024 |
| <ul style="list-style-type: none">Enhanced market intelligence visibility by building interactive Power BI dashboards, enabling near real-time KPI tracking and faster executive decision-making.Generated actionable insights by developing advanced Python queries to process large datasets, improving market research depth, and increasing analysis throughput by ~30%.Influenced strategic planning by presenting complex analytical findings to senior leadership, driving data-backed business initiatives. | | | |
| Technical Analyst | <u>Rochester Medical Center</u> | <i>Rochester, NY, USA</i> | 01/2024 - 06/2024 |
| <ul style="list-style-type: none">Improved EMR data accessibility by leading structured data collection initiatives, increasing research readiness, and reducing retrieval time by ~25%.Enabled <u>Lung Cancer Research</u> analytics by abstracting clinical data from Epic EMR and transcribing into REDCap, ensuring standardized and interoperable datasets.Increased study reliability by maintaining high-accuracy clinical abstraction and transcription, directly supporting downstream statistical analysis and research objectives. | | | |
| Senior Software Engineer | <u>Philips Healthcare</u> | <i>Bangalore, KA, India</i> | 01/2021 - 08/2023 |
| <ul style="list-style-type: none">Clinical Impact: Pioneered the design and deployment of a distributed and managed Clinical Data Lake for Novartis on AWS; provided end-to-end ownership for the Spinal Muscular Atrophy (SMA) clinical trials, delivering a compliant and secure data store for the <u>AI Tool Suite platform</u>.Full-Stack Engineering: Engineered the platform's core architecture by developing RESTful APIs and ReactJS frontend features; integrated PostgreSQL for scalable metadata storage and authored custom Terraform providers to automate secure, multi-tenant cloud deployments.System Architecture: Architected an automated, event-driven notification system for the <u>Clinical Data Lake</u>; leveraged Amazon SNS to decouple data ingestion from stakeholder communications, ensuring 100% notification reliability. | | | |

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- Cloud Optimization: Scaled the Clinical Data Lake by transforming the platform from a single-tenant to a secure multi-tenant architecture; optimized shared infrastructure, reducing customer overhead cost by ~35% while maintaining strict data isolation.
- DevOps & Infrastructure Modernization: Eliminated production downtime and manual intervention by migrating legacy Python deployment scripts to Terraform Infrastructure-as-Code; implemented blue-green deployment strategies.

Software Engineer [Philips Healthcare](#) *Bangalore, KA, India* **07/2019 - 01/2021**

- Delivered scalable backend services by developing Java Spring Boot microservices with RESTful APIs for a distributed Clinical Data Lake on Philips Health Suite Digital Platform (HSDP), improving system reliability and scalability.
- Advocated for a "Test-Driven" culture by authoring comprehensive unit test suites using Jest and React Testing Library for the frontend, and JUnit for Java Spring Boot backends; maintained over 70% code coverage for critical release features to ensure release stability.
- Standardized healthcare data interoperability by implementing ingestion pipelines compliant with HL7 FHIR STU3, enabling consistent clinical data modeling.
- Architected and implemented complex RBAC (Role-Based Access Control) workflows, including Create, Grant, and Revoke permission flows, ensuring secure data isolation for sensitive research study concepts.

Software Engineer Intern [Philips Healthcare](#) *Bangalore, KA, India* **08/2018 - 07/2019**

- Ensured regulatory compliance by localizing the [Philips Tasy EMR HTML5](#) system; developed and integrated SOAP/REST web services to implement customer-specific features aligned with local governance and audit requirements.
- Improved API reliability by building an end-to-end performance testing framework using Python and JMeter to load test 30+ APIs against SLAs, publishing results to engineering dashboards and Azure DevOps.

EDUCATION

Master of Science [University of Rochester](#) *Rochester, NY, USA* **08/2023 - 12/2024**

- Master of Science in Technical Entrepreneurship and Management
(Tools of Data Science, Marketing Management, Screening Technical Opportunities)

Master of Technology [VIT University](#) *Chennai, TN, India* **06/2014 - 04/2019**

- Master of Technology in Software Engineering
(Cloud Computing, Requirements Engineering, Software Architecture and Design)

ADDITIONAL INFORMATION

- Strengthened leadership engagement by serving as Core Representative of the Young Leaders Club at Philips, implementing reverse mentoring sessions and leading company-wide marketing initiatives.
- Drove community and academic collaboration by serving as VP of the Life Sciences Club and Project Manager for the Elevate Health Consulting Project with local Rochester businesses.
- Recognized consistently by architects and staff engineers for technical excellence and delivery across 5+ years at Philips.